

=====

MODULE 1: CELL BIOLOGY

1. (Multiple Choice) Which of the following transport mechanisms relies on the direct hydrolysis of ATP to move ions against their concentration gradient?
 - A. Voltage-gated sodium channels
 - B. Glucose/Sodium symporter
 - C. Sodium-Potassium (Na⁺/K⁺) pump
 - D. Aquaporins
 - E. Facilitated diffusion carriers
2. (True or False) The nucleolus is a membrane-bound organelle within the nucleus primarily responsible for the transcription of messenger RNA (mRNA).
3. (Open Question) Compare and contrast the cellular mechanisms of clathrin-mediated endocytosis and phagocytosis, including the types of cargo and the cytoskeletal elements involved.
4. (Fill in the Gap) In the fluid-mosaic model of the cell membrane, the sterol molecule that buffers membrane fluidity across different temperatures is _____.
5. (Matching) Match the cytoskeletal filament with its primary characteristic or component.
 6. Microtubules
 7. Actin filaments
 8. Intermediate filaments
 9. Nuclear lamins
 - A. Form a structural meshwork supporting the inner nuclear membrane.
 - B. Provide strong mechanical resistance to shear stress in epithelial tissues (e.g., keratins).
 - C. Composed of tubulin heterodimers; undergo dynamic instability.
 - D. Concentrated in the cell cortex; essential for cell crawling and cytokinesis.
10. (True or False Cluster) Evaluate the following statements regarding prokaryotic and eukaryotic cells:
 - A. Prokaryotes lack a membrane-enclosed nucleus.
 - B. Both prokaryotes and eukaryotes possess 80S ribosomes.
 - C. Eukaryotic DNA is linear and wrapped around histone proteins.
 - D. Prokaryotes typically divide via mitosis.
11. (Multiple Choice) The melting temperature [™] of a DNA double helix is primarily determined by its:
 - A. Length and the percentage of Adenine-Thymine (A-T) base pairs
 - B. Length and the percentage of Guanine-Cytosine (G-C) base pairs
 - C. Proportion of coding versus non-coding regions
 - D. Degree of histone acetylation
12. (Open Question) Describe the “fluid-mosaic model” of the plasma membrane, emphasizing the specific roles of cholesterol and integral membrane proteins in maintaining structural and functional integrity.
13. (Fill in the Gap) The macromolecular complex composed of snRNAs and proteins that removes introns from pre-mRNA is called the _____.
14. (Multiple Choice) Upon ligand binding, Receptor Tyrosine Kinases (RTKs) typically undergo which of the following immediate structural changes?
 - A. Uncoupling from a heterotrimeric G-protein
 - B. Opening of an intrinsic ion channel
 - C. Dimerization and autophosphorylation
 - D. Translocation directly to the nucleus

15. (True or False) G-Protein Coupled Receptors (GPCRs) are characterized by the presence of exactly seven transmembrane alpha-helical domains.
16. (Matching) Match the apoptotic protein with its cellular function.
17. Bcl-2
18. Bax
19. Caspase-9
20. Caspase-8
- A. Initiator caspase for the extrinsic apoptotic pathway.
 - B. Pro-apoptotic protein that forms pores in the outer mitochondrial membrane.
 - C. Anti-apoptotic protein that prevents cytochrome c release.
 - D. Initiator caspase activated within the apoptosome.
21. (Open Question) Explain the molecular basis of prion diseases, focusing on protein folding, secondary structural changes (alpha-helices vs beta-sheets), and the mechanism of aggregation.
22. (Fill in the Gap) In the CRISPR-Cas9 genome editing system, the Cas9 endonuclease requires a specific, short DNA sequence adjacent to the target site known as the _____ to bind and cleave the DNA.
23. (Multiple Choice) The G2/M cell cycle checkpoint prevents entry into mitosis primarily if:
- A. The cell lacks sufficient extracellular growth factors.
 - B. DNA replication is incomplete or DNA is significantly damaged.
 - C. Kinetochores are not properly attached to spindle microtubules.
 - D. The nucleolus has not fully disassembled.
24. (True or False Cluster) Evaluate the following statements regarding cyclins and CDKs:
- A. Cyclin-dependent kinases (CDKs) have constant concentration levels throughout the cell cycle.
 - B. Cyclins possess intrinsic kinase activity to phosphorylate targets.
 - C. The targeted degradation of cyclins via the proteasome ensures the cell cycle moves in one direction.
 - D. The Maturation Promoting Factor (MPF) is a complex of Cyclin B and CDK1.
25. (Multiple Choice) A mutation that results in the constitutive (constant) activation of the Ras GTPase protein is indicative of a:
- A. Gain-of-function mutation in a proto-oncogene
 - B. Loss-of-function mutation in a proto-oncogene
 - C. Gain-of-function mutation in a tumor suppressor gene
 - D. Loss-of-function mutation in a tumor suppressor gene
26. (Open Question) Discuss the application of patient-derived induced pluripotent stem cells (iPSCs) in modeling human genetic diseases in vitro, highlighting the specific transcription factors used for reprogramming.
27. (True or False) Telomerase is an RNA-dependent DNA polymerase that extends the 5' ends of linear chromosomes to prevent shortening during replication.
28. (Fill in the Gap) In Mendelian genetics, the phenomenon where a single genetic mutation produces multiple, seemingly unrelated phenotypic traits in an individual is known as _____.
29. (Multiple Choice) Which of the following features strictly differentiates an enveloped virus from a naked virus?
- A. The presence of a capsid
 - B. The possession of a lipid bilayer derived from the host cell
 - C. The ability to integrate into the host genome
 - D. The use of RNA as genetic material
30. (Open Question) Outline the process of co-translational translocation of a nascent polypeptide into the Rough Endoplasmic Reticulum. Be sure to mention the Signal Recognition Particle (SRP) and the translocon.
31. (Fill in the Gap) The cytogenetic technique that utilizes fluorescently labeled DNA probes to detect specific microdeletions or translocations on metaphase chromosomes is called _____.
32. (Multiple Choice) Eukaryotic general transcription factors (e.g., TFIID) typically bind to which highly conserved promoter sequence to initiate transcription?
- A. Shine-Dalgarno sequence
 - B. KDEL sequence

- C. TATA box
 - D. GC-rich island
33. (True or False) The small subunit of the eukaryotic ribosome is responsible for catalyzing the formation of the peptide bond during translation.
34. (Open Question) Differentiate between the concepts of penetrance and expressivity in Mendelian genetics, providing a brief conceptual definition for each.
35. (Multiple Choice) A point mutation in a DNA sequence that changes a codon specifying an amino acid into a premature STOP codon is known as a:
- A. Silent mutation
 - B. Missense mutation
 - C. Nonsense mutation
 - D. Frameshift mutation
36. (Matching) Match the signaling molecule or receptor to its location/type.
37. Steroid Hormone Receptor
38. cAMP
39. IP3
40. Adenylyl Cyclase
- A. Second messenger that releases calcium from the endoplasmic reticulum.
 - B. Intracellular transcription factor.
 - C. Intracellular second messenger generated from ATP.
 - D. Membrane-bound effector enzyme activated by G-alpha-s.
41. (Fill in the Gap) The degradative organelle characterized by an acidic lumen (pH ~5.0) and a high concentration of acid hydrolases is the _____.
42. (Open Question) Explain the concept of "contact inhibition" in normal somatic cells and describe how the loss of this property contributes to tumor progression.

=====

MODULE 2: HISTOLOGY

31. (Multiple Choice) Which type of intercellular junction is primarily responsible for sealing the paracellular space between adjacent epithelial cells, preventing the free passage of luminal contents?
- A. Desmosome (Macula adherens)
 - B. Gap junction
 - C. Tight junction (Zonula occludens)
 - D. Hemidesmosome
32. (Open Question) Differentiate between the three modes of exocrine gland secretion (merocrine, apocrine, and holocrine) and identify one specific anatomical gland that utilizes the holocrine method.
33. (Multiple Choice) The extracellular matrix of proper connective tissue is composed of fibrous proteins and ground substance. Which of the following is a primary component of the ground substance?
- A. Type I Collagen
 - B. Elastin
 - C. Glycosaminoglycans (GAGs)
 - D. Reticular fibers
34. (Fill in the Gap) The concentric layers of calcified bone matrix surrounding the central canal of an osteon are referred to as _____.
35. (Matching) Match the specialized cell of the skeletal system to its function or characteristic.
36. Osteoblast
37. Osteoclast

38. Osteocyte
39. Chondrocyte
- A. Mature bone cell residing within a lacuna.
 - B. Multinucleated giant cell responsible for bone resorption.
 - C. Resides in avascular tissue; maintains the cartilage matrix.
 - D. Synthesizes and secretes unmineralized osteoid.
40. (True or False Cluster) Evaluate the following statements regarding muscle tissue:
- A. Skeletal muscle fibers are multinucleated syncytia resulting from the fusion of myoblasts.
 - B. Cardiac muscle cells are mechanically coupled by intercalated discs.
 - C. Smooth muscle cells contain visible striations under light microscopy.
 - D. Smooth muscle contraction is regulated by calcium binding to calmodulin.
41. (Multiple Choice) Which lymphoid organ is uniquely characterized by the presence of a cortex and medulla, lobules, and specialized epithelial reticular cells that form Hassall's corpuscles?
- A. Spleen
 - B. Lymph node
 - C. Thymus
 - D. Palatine tonsil
42. (Open Question) Describe the histological organization of the osteon (Haversian system) in compact bone. Detail the roles of the central canal, lamellae, lacunae, and canaliculi.
43. (Fill in the Gap) In the soma of a multipolar neuron, the intensely basophilic clumps of rough endoplasmic reticulum visible under light microscopy are called _____ bodies.
44. (Multiple Choice) Which glial cell is responsible for producing the myelin sheath around axons within the Central Nervous System (CNS)?
- A. Schwann cell
 - B. Astrocyte
 - C. Oligodendrocyte
 - D. Microglia
45. (True or False) Brown adipose tissue generates heat because its mitochondria express Uncoupling Protein 1 (Thermogenin), which bypasses ATP synthase.
46. (Multiple Choice) The respiratory tract (e.g., trachea) is primarily lined by which specific type of surface epithelium?
- A. Simple squamous epithelium
 - B. Stratified squamous non-keratinized epithelium
 - C. Pseudostratified ciliated columnar epithelium
 - D. Transitional epithelium
47. (Open Question) Compare the histological appearance and primary physiological roles of white adipose tissue versus brown adipose tissue.
48. (Fill in the Gap) The fundamental contractile unit of a skeletal muscle fiber, spanning from one Z-line to the adjacent Z-line, is the _____.
49. (Multiple Choice) Tendons and ligaments are characterized by tightly packed, parallel bundles of collagen fibers. This tissue type is classified as:
- A. Loose areolar connective tissue
 - B. Dense irregular connective tissue
 - C. Dense regular connective tissue
 - D. Reticular connective tissue
50. (True or False) Endocrine glands lack a duct system and secrete their products (hormones) directly into the surrounding capillary networks.
51. (Open Question) Describe the specialized structure of the neuromuscular junction (motor end plate), identifying the pre-synaptic features, the synaptic cleft, and the post-synaptic specializations of the sarcolemma.
52. (Fill in the Gap) In the blood-brain barrier, the continuous tight junctions of the capillary endothelial cells are structurally supported and maintained by the perivascular end-feet of _____.
53. (Multiple Choice) In a standard Hematoxylin and Eosin (H&E) stain, the cytoplasm of an erythrocyte stains intensely pink/red (eosinophilic) primarily due to the high concentration of:

- A. RNA
- B. Hemoglobin
- C. DNA
- D. Lipids

54. (True or False Cluster) Evaluate the following statements regarding lymph nodes:
- A. Afferent lymphatic vessels enter the lymph node at the convex surface.
 - B. The paracortex is primarily populated by T-lymphocytes.
 - C. The superficial cortex contains lymphoid follicles/nodules rich in B-lymphocytes.
 - D. Efferent lymphatic vessels exit the lymph node at the hilum.
55. (Open Question) Outline the path of a senescent red blood cell traversing the splenic red pulp. Explain the structural basis (e.g., sinusoids and macrophage cords) of the spleen's filtering function.
56. (Multiple Choice) Blood platelets are not true cells, but rather anucleate cytoplasmic fragments derived from which giant bone marrow precursor cell?
- A. Reticulocyte
 - B. Macrophage
 - C. Megakaryocyte
 - D. Promyelocyte
57. (Multiple Choice) Osteoporosis is characterized by a pathological decrease in bone mass. Histologically, this is primarily driven by the overactivity of which specific cell type relative to its counterpart?
- A. Osteoblast
 - B. Osteoclast
 - C. Osteocyte
 - D. Chondroblast
58. (True or False) The stratum basale of the epidermis consists of a single layer of mitotically active stem cells resting on the basement membrane.

=====

MODULE 3: EMBRYOLOGY

55. (Multiple Choice) During the first week of development, the solid ball of 16-32 blastomeres formed by cleavage is known as the:
- A. Zygote
 - B. Morula
 - C. Blastocyst
 - D. Gastrula
56. (True or False) In the female ovarian cycle, the luteal phase is characterized by the secretion of progesterone from the corpus luteum to maintain the uterine lining.
57. (Open Question) Describe the role of the notochord during the third week of development, particularly focusing on its structural fate and its inductive effect on the overlying embryonic layers.
58. (Fill in the Gap) The fetal portion of the placenta is primarily derived from the highly vascularized _____ frondosum.
59. (Matching) Match the severe developmental defect with its embryological cause.
60. Spina bifida
61. Anencephaly
62. Gastroschisis
63. Sirenomelia
- A. Failure of the cranial neuropore to close.
 - B. Defective formation of the caudal mesoderm (caudal dysgenesis).
 - C. Failure of the caudal neuropore to close.
 - D. Defective closure of the lateral body wall folds, leading to extruded intestines.

64. (Multiple Choice) Which embryonic germ layer gives rise to the epithelial lining of the gastrointestinal and respiratory tracts?
- Ectoderm
 - Mesoderm
 - Endoderm
 - Neural Crest
65. (Open Question) Explain the hormonal regulation of the ovarian cycle, detailing the specific roles of LH (Luteinizing Hormone) and FSH (Follicle-Stimulating Hormone) in follicle maturation and ovulation.
66. (True or False) The amniotic cavity forms strictly within the hypoblast layer of the bilaminar embryonic disc.
67. (Multiple Choice) Somites are distinct blocks of mesoderm that sequentially develop along the neural tube. They originate from which specific region of the mesoderm?
- Lateral plate mesoderm
 - Intermediate mesoderm
 - Paraxial mesoderm
 - Extraembryonic mesoderm
68. (Fill in the Gap) During blastocyst formation, the outer cell layer that will contribute to the placenta is the trophoblast, while the inner cluster of cells that will form the embryo proper is called the _____.
69. (Open Question) Describe the formation of the primitive streak during human gastrulation. Explain its role in establishing the body axes and facilitating the generation of the trilaminar disc.
70. (Multiple Choice) The process by which spermatozoa undergo a biochemical conditioning in the female reproductive tract, allowing them to undergo the acrosome reaction, is called:
- Capacitation
 - Cleavage
 - Compaction
 - Cortical reaction

MODULE 4: INTERDISCIPLINARY

67. (Multiple Choice) Retroviruses like HIV exhibit targeted viral tropism. Which specific cell surface receptor dictates the entry of HIV into T-lymphocytes, heavily compromising the adaptive immune system?
- CD8
 - CD4
 - MHC Class I
 - Toll-like receptor 4 (TLR4)
68. (True or False) Kartagener syndrome (primary ciliary dyskinesia) is caused by a structural defect in the dynein arms of cilia, often resulting in both chronic respiratory infections and embryonic situs inversus.
69. (Open Question) Parkinson's Disease involves progressive neurodegeneration. Identify the specific histological brain region and the neurotransmitter-producing neuron type that are lost. Briefly explain how stem cell technologies (like iPSCs) can be utilized to model this disease in vitro.
70. (Fill in the Gap) Osteogenesis imperfecta is primarily caused by genetic mutations affecting the folding and assembly of Type _____ collagen, leading to structurally compromised bone matrix.

ANSWER KEY & RUBRIC

- C (Na⁺/K⁺ pump requires ATP for primary active transport)
- False (The nucleolus is not membrane-bound and synthesizes rRNA, not mRNA)
- [Rubric - OQ]: Clathrin-mediated endocytosis involves specific receptor-ligand complexes coated by clathrin to form small vesicles. Phagocytosis is the engulfment of large particles (bacteria/debris) driven by actin microfilament rearrangements to form large phagosomes.
- Cholesterol
- 1-C, 2-D, 3-B, 4-A
- A-True, B-False, C-True, D-False

7. B (G-C pairs have 3 hydrogen bonds, requiring higher temperature to denature than A-T pairs which have 2)
8. [Rubric - OQ]: The membrane is a fluid bilayer of phospholipids. Cholesterol buffers fluidity (prevents tight packing in cold, restricts movement in heat). Integral proteins span the membrane serving as transporters, receptors, or anchors.
9. Spliceosome
10. C (Ligand binding induces RTK dimerization and cross-autophosphorylation of intracellular tyrosine residues)
11. True (GPCRs are 7-transmembrane receptors)
12. 1-C, 2-B, 3-D, 4-A
13. [Rubric - OQ]: Prion diseases occur when the normal PrP^C protein (alpha-helix rich) misfolds into the pathogenic PrP^{Sc} conformation (beta-sheet rich). This misfolded state catalyzes the misfolding of normal proteins, leading to amyloid plaque aggregation and neurodegeneration.
14. PAM (Protospacer Adjacent Motif)
15. B (The G2/M checkpoint ensures DNA replication is complete and undamaged)
16. A-True, B-False, C-True, D-True
17. A (Ras is a proto-oncogene; constitutive activation promotes uncontrolled cell growth)
18. [Rubric - OQ]: Somatic cells (e.g., fibroblasts) are reprogrammed into iPSCs using Yamanaka factors (Oct4, Sox2, c-Myc, Klf4). These iPSCs can be differentiated into the specific cell type affected by the patient's genetic disease (e.g., neurons) to study pathophysiology or test drugs in vitro.
19. False (Telomerase extends the 3' overhangs, not the 5' ends)
20. Pleiotropy
21. B (Enveloped viruses possess a host-derived lipid bilayer; naked viruses do not)
22. [Rubric - OQ]: A signal sequence on the nascent polypeptide is recognized by the SRP. The SRP pauses translation and binds to an SRP receptor on the ER membrane. The ribosome docks at the translocon pore, translation resumes, and the protein is threaded into the ER lumen.
23. FISH (Fluorescence In Situ Hybridization)
24. C (The TATA box is bound by TBP, a subunit of TFIID)
25. False (The large subunit contains the peptidyl transferase ribozyme activity)
26. [Rubric - OQ]: Penetrance refers to the percentage of individuals with a specific genotype who actually express the associated phenotype (all-or-nothing). Expressivity refers to the degree or severity to which the phenotype is expressed among individuals with the genotype.
27. C (Nonsense mutation creates a premature stop codon)
28. 1-B, 2-C, 3-A, 4-D
29. Lysosome
30. [Rubric - OQ]: Normal somatic cells halt proliferation when they form a confluent monolayer and touch neighboring cells. Tumor cells lose contact inhibition, allowing them to pile up, overgrow, and eventually invade surrounding tissues (metastasis).
31. C (Tight junctions / Zonula occludens seal the paracellular space)
32. [Rubric - OQ]: Merocrine: exocytosis of vesicles (e.g., salivary). Apocrine: pinching off of apical cytoplasm (e.g., lactating mammary). Holocrine: entire cell ruptures to release product (e.g., sebaceous gland).
33. C (GAGs are the primary carbohydrate component of ground substance)
34. Lamellae
35. 1-D, 2-B, 3-A, 4-C
36. A-True, B-True, C-False, D-True
37. C (Thymus contains Hassall's corpuscles)
38. [Rubric - OQ]: The osteon consists of a central Haversian canal containing blood vessels/nerves. It is surrounded by concentric lamellae of bone matrix. Osteocytes reside in lacunae between lamellae and communicate with the canal and each other via tiny channels called canaliculi.
39. Nissl
40. C (Oligodendrocytes myelinate the CNS; Schwann cells myelinate the PNS)
41. True (Thermogenin uncouples the proton gradient to produce heat)
42. C (Pseudostratified ciliated columnar epithelium)
43. [Rubric - OQ]: White adipose is unilocular (one large lipid droplet), pushing the nucleus to the periphery; it functions in energy storage. Brown adipose is multilocular (many small droplets) with central nuclei and abundant mitochondria; it functions in non-shivering thermogenesis.
44. Sarcomere
45. C (Dense regular connective tissue)
46. True (Endocrine glands are ductless)
47. [Rubric - OQ]: Pre-synaptic: axon terminal containing vesicles of acetylcholine. Synaptic cleft: extracellular gap where neurotransmitter is released. Post-synaptic: highly folded sarcolemma containing acetylcholine receptors.
48. Astrocytes
49. B (Hemoglobin is a basic protein that heavily binds the acidic dye eosin)

50. A-True, B-True, C-True, D-True
51. [Rubric - OQ]: RBCs leave the open circulation of splenic cords (cords of Billroth) and must squeeze between the stave cells lining the venous sinusoids to re-enter circulation. Stiff, old, or damaged RBCs cannot deform enough to pass and are phagocytosed by resident macrophages.
52. C (Megakaryocytes fragment their cytoplasm to form platelets)
53. B (Osteoclasts resorb bone)
54. True (The stratum basale is the deepest, mitotically active layer)
55. B (Morula is the 16-32 cell stage)
56. True (The corpus luteum secretes progesterone to prepare/maintain the endometrium)
57. [Rubric - OQ]: The notochord forms a midline axis. It induces the overlying ectoderm to thicken and form the neural plate (neurulation). Structurally, it ultimately degenerates but persists as the nucleus pulposus of the intervertebral discs.
58. Chorion
59. 1-C, 2-A, 3-D, 4-B
60. C (Endoderm lines the GI and respiratory tracts)
61. [Rubric - OQ]: FSH stimulates the growth and maturation of ovarian follicles. An estrogen peak from the mature follicle triggers a massive LH surge from the anterior pituitary, which causes the follicle to rupture and release the secondary oocyte (ovulation).
62. False (The amniotic cavity forms within the epiblast layer)
63. C (Paraxial mesoderm forms somites)
64. Inner Cell Mass (or Embryoblast)
65. [Rubric - OQ]: The primitive streak appears on the surface of the epiblast, establishing the cranial-caudal and left-right body axes. Epiblast cells invaginate through the streak, displacing the hypoblast to form endoderm and filling the space to form mesoderm, converting the bilaminar disc to a trilaminar disc.
66. A (Capacitation removes glycoprotein coats from the sperm plasma membrane)
67. B (HIV uses the CD4 receptor to infect T-helper cells)
68. True (Defective cilia disrupt both mucociliary clearance and embryonic left-right axis determination)
69. [Rubric - OQ]: Disease involves the loss of dopaminergic neurons in the substantia nigra (midbrain). Patient-derived somatic cells can be reprogrammed into iPSCs and directed to differentiate into dopaminergic neurons, allowing researchers to study the disease pathology and screen drugs in a dish.
70. I (Type I collagen)